

Week 7; Data Cleaning and Preprocessing Fundamentals

Learning Objectives

- Understand why data cleaning is essential for data science and AI model development
- Identify common issues with raw data including missing values and outliers
- Apply fundamental data cleaning and preprocessing techniques using real-world datasets

Agenda

1. Introduction to Data Cleaning; Why it Matters
2. Handling Missing Data; Approaches and Best Practice
3. Detecting and Treating Outliers
4. Basic Data Preprocessing Steps and Transformations
5. Industry Example; Preparing AI Training Data
6. Hands-On Activity; Cleaning Sample Data
7. Recap and Assessment Preparation

Why Data Cleaning is Critical

- Raw data is rarely ready for analysis or machine learning; cleaning is needed to avoid misleading results
- Quality cleaning reduces errors, improves AI performance, and builds trust in results
- Large datasets used in AI (e.g. LAION-5B, COCO) must be cleaned for scalability and fairness

Common Data Problems in Large Datasets

- Missing values; data entries are blank or incomplete
- Outliers; data points that differ greatly from others
- Inconsistent formatting; non-standard or mixed input formats
- Duplicated entries; repeated records can bias results
- Irrelevant or redundant features; unnecessary data complicates analysis

Handling Missing Data

- Deletion; remove rows or columns with lots of missing values
- Imputation; fill missing values with mean, median, or mode
- Model-based imputation; predict missing values using available data
- Industry tip; always check how missing values might affect AI model fairness or bias

Code Example; Finding and Treating Missing Data (pandas)

```
import pandas as pd

# Load data
df = pd.read_csv('dataset.csv')

# Identify missing values
print(df.isnull().sum())

# Simple imputation
df['age'].fillna(df['age'].mean(), inplace=True)

# Remove rows with any missing values
df.dropna(inplace=True)
```

- Choose approach based on dataset size, context, and impact

Detecting and Dealing with Outliers

- Outliers may result from errors or represent rare real-world events
- Simple detection; visualize with boxplots or scatterplots, use z-score or IQR calculations
- Treatments; remove, replace, or transform outliers depending on context
- Always document decisions about outliers for reproducibility

Example; Outlier Detection and Treatment

```
import numpy as np

# Boxplot for 'income' column
df.boxplot(column='income')

# Remove data points outside 3 standard deviations
df = df[(np.abs(df['income'] - df['income'].mean()) / df['income'].std()) < 3]
```

- Visual inspection helps catch data entry errors or rare cases

Basic Data Preprocessing Steps

- Convert data types; e.g. change strings to numerical values
- Normalize or scale values for ML algorithms
- Encode categorical data into numbers (e.g. one-hot encoding)
- Split data for training and testing to validate models
- Consistent preprocessing ensures fair, reliable AI outcomes

Industry Context; Cleaning Large AI Datasets

- Example; LAION-5B images must be checked for duplicates, corrupted files, and missing labels
- Large-scale AI projects require automated pipelines for cleaning; manual work is not scalable
- Tools like HuggingFace Datasets and open source Python libraries support reproducible preprocessing
- Documentation of all cleaning steps is critical for compliance and transparency

Activity; Hands-On Data Cleaning

- In groups, use a Jupyter Notebook to:
 - Load a sample open dataset (from UCI ML Repo or similar)
 - Identify missing values and decide whether to remove or impute
 - Detect outliers using boxplots or z-score method
 - Apply at least one transformation (e.g. normalization)
 - Document your cleaning process
- Discuss; How might these steps differ with a much larger or more complex AI dataset?

Assessment Preparation; What You Need to Demonstrate

- Ability to identify and treat missing data and outliers using Python tools
- Clear documentation of cleaning decisions and rationales
- Awareness of data ethics and impact of cleaning on AI models
- Link back to previous concepts; data quality, bias, and documentation from Weeks 5 and 6

Recap and Next Steps

- Data cleaning is a foundational skill for reliable analysis in data science and AI
- Effective preprocessing makes large datasets usable and trustworthy for model training
- In next week; Learn about versioning and documenting your data cleaning workflows

Further Reading and Resources

- "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron; Chapters on data cleaning and preprocessing
- UCI Machine Learning Repository; Try more datasets for practice
- HuggingFace Datasets documentation; Building scalable cleaning pipelines

